Kalva Siri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.69
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education (BIE), Telangana)	2024	97.80%
Sri Chaitanya School	Class X (Board of Secondary Education (BSE), Telangana)	2022	10.00%

SKILLS

Programming Languages:

C, C++, Python, HTML, CSS, JavaScript, SQL.

Tools & Libraries:

NumPy, Pandas, Matplotlib, MATLAB, PostgreSQL, LATEX, Git, GitHub.

Soft Skills:

Analytical Thinking, Problem Solving, Communication Skills, Presentation Skills, Team collaboration.

RELEVANT COURSES

Mathematics:

Linear Algebra, Probability Theory, Applied Statistics, Convex Optimization, Ordinary Differential Equations, Calculus - I, II, Introduction to Number Theory.

Computer Science:

Introduction to Programming, Data Structures and Applications, Algorithms, Theory of Computation, Compilers - I, Database Management Systems - I.

PROJECTS

Nearest Neighbor Search using K-D trees | DSA Lab project | Drive link

(Data Structures & Algorithms, Computational Geometry, Algorithm Optimization, Complexity Analysis, C++, Latex)

- Built a multi-dimensional tree to perform efficient radius-based nearest searches in a multi-dimensional (k-dimensional) space.
- Implemented recursive tree construction, multidimensional range search, and branch pruning for optimized query performance.
- Generated random datasets to evaluate correctness and compare performance against brute-force search.
- Analyzed algorithmic complexity and scalability for high-dimensional spatial data.
- Compared KD trees with alternative nearest-neighbor search techniques including : **Generalized Hyperplane trees (GHT), Vantage-Point (VP) trees, Locality-Sensitive Hashing (LSH).**

Applied Statistics and Data Visualization - Python | Applied Statistics project | Drive link

(Python, NumPy, Pandas, Matplotlib, ReportLab, Descriptive Statistics, Data Visualization, Statistical Inference)

- Performed statistical analysis on real-world grouped and ungrouped datasets using **Python, NumPy, Pandas, Matplotlib.**
- Generated and interpreted visualizations including **frequency tables, bar charts, histograms, relative frequency histograms, frequency polygons, pie charts, and box plots.**
- Applied statistical inference techniques such as **sampling distributions, point estimation, interval estimation, and hypothesis testing.**
- Constructed and Analyzed **2x2 contingency tables** to study relationships between categorical variables.
- Interpreted statistical results to derive meaningful insights from data and support data-driven decision-making.

SCHOLASTIC ACHIEVEMENTS

- Achieved a **top 8%** rank in Stage 1 and **top 5.75%** rank in Stage 2 among **250,000+ applicants** during the **AlgoUniversity Technology Fellowship (ATF) 2025** selection process, qualifying to apply for a 70% scholarship. Certificate

- Achieved a maximum rating of **1437 (2 stars)** in codechef. profile

POSITIONS OF RESPONSIBILITY

Coordinator - Social Media Handling Team, Indian Knowledge Systems (IKS) Student's Chapter.

- Currently managing the official website of IKS Student's Chapter.
- Actively participating in field research and discussions.